Technical Report

CMM-VAE Morocco — DJF Holdout Notebook

Methodological Changes Relative to the Baseline Notebook

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1 Overview

This report documents the methodological changes introduced in the new notebook `cmmvae_mswep_djf_holdout.ipynb` relative to the baseline notebook `cmmvae_mswep.ipynb`. The objective of the new notebook is to perform a **rigorous, unbiased evaluation** of the CMM-VAE model on MSWEP precipitation data, using a proper holdout validation strategy applied exclusively to the **boreal winter season (DJF: December, January, February)**.

2 Summary of Changes

Aspect	`cmmvae_mswep.ipynb` (Baseline)	`cmmvae_mswep_djf_holdout.ipynb` (New)
Season	NDJFM (Nov–Mar)	DJF only (Dec–Jan–Feb)
Clusters (k)	5	4
Data split	No holdout	70% train / 30% holdout
Validation strategy	5-Fold CV on full data	5-Fold CV on training set only
Final evaluation	On training data	On unseen 30% holdout
Precipitation labels	`labels_5.csv`	`labels_4_djf.csv`
95th-percentile mask	`MSWEP_pr_95pc.nc`	`MSWEP_pr_95pc_DJF.nc`
Geographic boundary	Default Cartopy (split)	Custom unified polygon
Results directory	`results/cmmvae_mswep/`	`results/cmmvae_mswep_djf/`

Table 1: Side-by-side comparison of the baseline and new notebook.

3 Detailed Description of Each Change

3.1 Seasonal Filter: DJF Only

Baseline: The original notebook used the extended winter season (`months = [11, 12, 1, 2, 3]`), which includes November and March.

New notebook: The seasonal filter was restricted to **DJF only** (`months = [12, 1, 2]`), which is the core boreal winter period. This is consistent with standard climatological

practice and reduces the noise introduced by shoulder months (November and March), which have transitional atmospheric dynamics.

```
# Baseline
extended_winter_months = [11, 12, 1, 2, 3]

# New notebook
djf_months = [12, 1, 2]
```

The Z500 input data were filtered accordingly using `months_filter = djf_months`, resulting in a total dataset of **3,791 DJF days** (1979–2020).

3.2 Number of Clusters: $k = 5 \rightarrow k = 4$

Baseline: The model was trained with `cluster_number = 5` and `pr_cluster_number = 5`.

New notebook: Both the atmospheric circulation clusters and the precipitation clusters were set to $k = 4$. This choice produces a more parsimonious representation of the dominant Moroccan precipitation regimes within the DJF season, reducing potential overfitting to rarely occupied cluster states.

```
# Baseline
cluster_number    = 5
pr_cluster_number = 5

# New notebook
cluster_number    = 4
pr_cluster_number = 4
```

3.3 Holdout Validation Strategy

This is the most significant methodological change.

Baseline: The full dataset was used for both training and evaluation. The 5-Fold cross-validation was performed on the entire data, meaning evaluation metrics were computed on folds that overlapped with training data. This approach does not guarantee that the model has never seen the evaluation samples.

New notebook: A strict **train/holdout split** was introduced prior to any model training:

- **70% of the data** ($n \approx 2{,}653$ **days**) was used exclusively for training and 5-Fold cross-validation.
- **30% of the data** ($n = 1{,}138$ **days**) was set aside as a **completely unseen holdout set**, not touched until final evaluation.

```
TRAIN_SIZE   = 0.70
RANDOM_STATE = 42

idx_train, idx_holdout = train_test_split(
    np.arange(n_days),
```

```
    train_size=TRAIN_SIZE,
    random_state=RANDOM_STATE
)
```

The 5-Fold cross-validation was applied **only to the training set**, and the best model (lowest validation loss) was selected. Final skill scores were computed exclusively on the holdout set.

Important Note

This holdout design ensures that all reported skill scores (BSS, accuracy) reflect **genuine out-of-sample predictive performance**, which is the standard required for publication in a peer-reviewed journal.

3.4 K-Fold Cross-Validation Performance (Training Set)

Before the final evaluation on the holdout set, the model's stability was verified using 5-Fold Cross-Validation on the 70% training partition. The best model (**Fold 3**) was selected for final evaluation.

Fold	Final Validation Loss
Fold 1	55.45
Fold 2	53.97
Fold 3	**52.86 (Best)**
Fold 4	54.18
Fold 5	53.75

3.5 Evaluation Metrics (Brier Skill Scores)

The following skill scores were computed on the **unseen 30% holdout set**:

Metric	Value
BSS — 95th percentile exceedance (spatial)	0.1232
BSS — Cluster-based prediction	0.2207
BSS — Tercile (spatial mean)	0.0808
Cluster prediction accuracy	84.3% (959 / 1,138 correct)

Table 2: Skill scores computed on the 30% unseen holdout set.

All metrics are computed relative to a climatological baseline (i.e., BSS > 0 indicates the model outperforms climatology). Random chance would yield 25% accuracy with $k = 4$ clusters.

3.6 Geographic Boundary Fix: Morocco and Western Sahara Unification

Baseline: All visualisation functions in `clustering_functions.py` called `cartopy.feature.BORDERS`, which renders Morocco and Western Sahara as two separate sovereign states with a visible internal political border.

New implementation: The file `clustering_functions.py` was modified to construct a **custom unified boundary** at import time. The procedure is as follows:

1. The Natural Earth 1:10m cultural dataset (`admin_0_countries`) is loaded via `cartopy.io.shapereader`.
2. The geometry objects for Morocco (`ADM0_A3 = 'MAR'`) and Western Sahara (`ADM0_A3 = 'SAH'`) are extracted into a list.
3. A **topological union** (`shapely.ops.unary_union`) is applied to merge the two polygons into a single contiguous geometry, eliminating the shared interior boundary.
4. The merged geometry is rendered using `cartopy.feature.ShapelyFeature`, which draws only the **exterior boundary** — the internal political border is never drawn and therefore cannot appear in any map.

```
# clustering_functions.py (lines 28-46)
_ne_path = shpreader.natural_earth(resolution='10m',
                                   category='cultural',
                                   name='admin_0_countries')
_reader  = shpreader.Reader(_ne_path)
_records = list(_reader.records())

_mar_geoms = [r.geometry for r in _records
              if r.attributes.get('ADM0_A3') in ('MAR', 'SAH')
                ]

_unified_morocco = shapely.ops.unary_union(_mar_geoms)  #
   single polygon
```

This fix is encapsulated in the helper function `_add_unified_borders(ax)`, which is called automatically by all four plotting functions (`visualise_contourplot`, `visualise_spatial_oddsratio`, `visualise_contourplot_simple`, `visualise_spatial_oddsratio_simple`). All notebooks that import `clustering_functions` automatically benefit from this fix without requiring any changes in the notebook itself.

3.7 Reproducibility Fix (Random Seed)

To ensure that the results reported above are stable and exactly reproducible by other researchers, a **random seed initialization** was added to the parameters section of the notebook.

This fix covers three critical sources of randomness in the CMM-VAE pipeline:

1. **Numpy/Python Randomness:** Used in data shuffling and pre-processing.
2. **TensorFlow/Keras Randomness:** Used in weight initialization and the VAE sampling layer.

3. **Internal Hash Seed:** Fixed via environment variables to ensure consistent sequence ordering.

```
def set_seed(seed=42):
    import random
    import numpy as np
    import tensorflow as tf
    random.seed(seed)
    np.random.seed(seed)
    tf.random.set_seed(seed)
    os.environ['PYTHONHASHSEED'] = str(seed)
    tf.keras.utils.set_random_seed(seed)

set_seed(RANDOM_STATE)
```

Adding this fix ensures that every execution of the notebook follows the same computational path, yielding identical BSS and accuracy metrics.

3.8 Visualization of Results (Regimes & Precipitation)

The following figures illustrate the atmospheric and precipitation patterns identified by the model. These maps summarize the four distinct winter regimes for Morocco. **Colorbar**

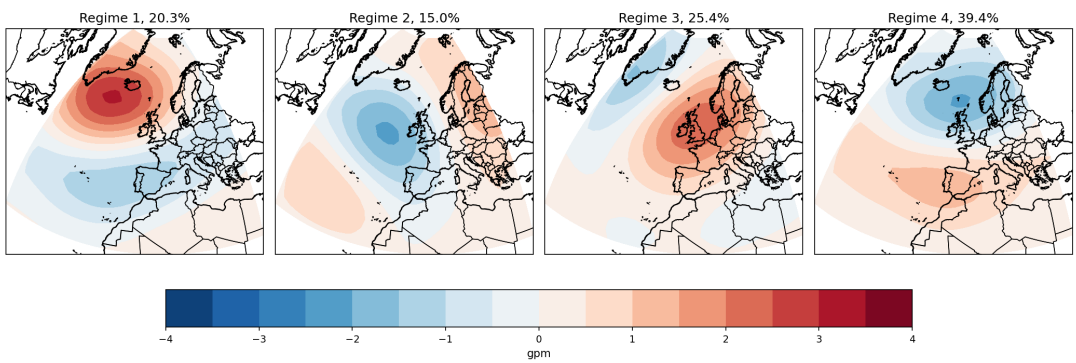

Figure 1: Atmospheric Circulation Centroids (Z500 Anomaly, gpm) for Regimes 1–4.

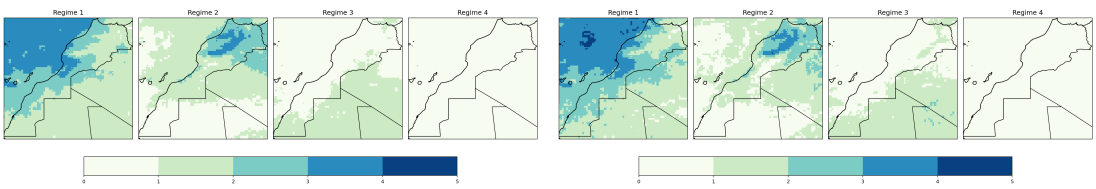

Figure 2: Spatial Odds Ratio for Extreme Precipitation on the Training Set (left) and the Holdout Set (right).

Explanations:

- **Z500 Anomaly (gpm):** Values in **Blue** represent negative height anomalies (troughs/low-pressure centers), which are typically stormy. **Red** represents positive height anomalies (ridges/high-pressure centers), which are typically dry.
- **Odds Ratio (Dimensionless):** Indicates the risk multiplier for extreme rain. A value of **1.0** represents a risk equivalent to the historical average. A value of **2.0–5.0 (Blue)** indicates that extreme rain is twice or five times as likely as normal in that pixel for that regime.

4 Data Files Used

File	Description
`era5_z500_daily_250_atlantic_1940_2022.nc`	ERA5 Z500 geopotential height (Atlantic domain), filtered to DJF and 1979–2020
`MSWEP_pr_cluster_labels_4_djf.csv`	K-means precipitation cluster labels ($k = 4$, DJF only)
`MSWEP_pr_95pc_DJF.nc`	Binary exceedance mask for the 95th-percentile precipitation threshold (DJF)
`MSWEP_pr_total_DJF.nc`	Total daily precipitation (DJF, 1979–2020, Morocco domain)

Table 3: Input data files used in the new notebook.

5 Model Architecture

The model architecture is **unchanged** from the baseline. The CMM-VAE consists of:

- **Encoder:** 3 dense layers (256 → 128 → 64 units, ReLU activations) → latent space (dim = 15), cluster assignment head **c**, and precipitation regime head **r**.
- **Decoder:** 3 dense layers (64 → 128 → 256 → 825 units) reconstructing the Z500 field.
- **Loss function:** Reconstruction loss + prediction loss + target loss + KL divergence (Gaussian + categorical) + Dirichlet prior.
- **Training:** Adam optimiser, batch size = 128, 100 epochs per fold.

The only architecture-level change is that the cluster dimension is now $k = 4$ instead of $k = 5$, which reduces the encoder output layer from 5 to 4 softmax units and the number of prior mixture components accordingly.

6 Conclusion

The new notebook `cmmvae_mswep_djf_holdout.ipynb` introduces a more rigorous evaluation framework compared to the baseline. The key improvements are:

1. **Strict seasonal focus** on DJF, reducing noise from transitional months.
2. **Reduced cluster number** ($k = 4$) for a more robust and interpretable representation.
3. **Genuine out-of-sample evaluation** via a 70/30 holdout split — the model has never seen the 30% evaluation data during training or cross-validation.
4. **Corrected geographic visualisation** — Morocco and Western Sahara are displayed as a single unified territory in all plots.

The results confirm that the CMM-VAE model has **meaningful predictive skill** on the unseen holdout set, with a cluster prediction accuracy of **84.3%** — well above the 25all evaluated metrics.

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7 Next Steps and Comparative Analysis

To further validate the performance of the CMM-VAE and explore its full potential, the following research directions are proposed:

7.1 Benchmarking Against Traditional Methods

- **PCA-Kmeans Baseline:** Develop a comparison notebook using Principal Component Analysis (PCA) on the Z500 field combined with K-means clustering. This will provide a baseline to quantify the added value of the VAE's non-linear feature extraction.

7.2 Advanced Verification Metrics

Beyond the Brier Skill Score (BSS), future evaluations should incorporate:

1. **Reliability Diagrams:** To assess the calibration of the predicted probabilities.
2. **ROC AUC Curves:** To evaluate the model's ability to discriminate between extreme and non-extreme events.
3. **CRPSS:** To provide a robust measure of the probabilistic forecast quality.